

CHE 301: Thermodynamics

Assignment 2

Submitted by

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Abstract

This report presents a thermodynamic analysis of a regenerative Rankine cycle with a closed feedwater heater (CFWH). The study develops steady-state mass, energy, and entropy balances for each component (boiler, HP/LP turbines, condenser, pump, CFWH), evaluates states using the CHE calculator, and applies isentropic and efficiency relations to obtain reversible and actual performance. Key outcomes include the steam requirement, fuel consumption, turbine and cycle efficiencies, component-wise lost work (irreversibilities), and associated CO₂ emissions. The results are presented through a state table, component analyses, and a complete energy and exergy balance, highlighting performance limitations.

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1 Problem Statement

Superheated steam is generated in a boiler at $P = 80 \text{ bar}$ and $T = 480^\circ\text{C}$ and expanded in a high-pressure (HP) turbine to 15 bar. A fraction of the flow (y) is extracted to a closed feedwater heater (CFWH) at 15 bar, while the remaining $(1 - y)$ fraction expands further in a low-pressure (LP) turbine to $P = 8 \text{ kPa}$. The condenser cools the exhaust to subcooled liquid at $T = 38^\circ\text{C}$. The feedwater is then pumped, heated in the CFWH, and returned to the boiler. Boiler efficiency is 80%, pump efficiency is 70%, and turbine efficiencies are 80%.

The goal is to calculate:

1. Required steam flow (MMSCFD).
2. Coal consumption rate (kg/s).
3. Lost work in each component.
4. Cycle thermal and second-law efficiencies.
5. Annual CO_2 emissions (Gt/year).

2 System Schematic

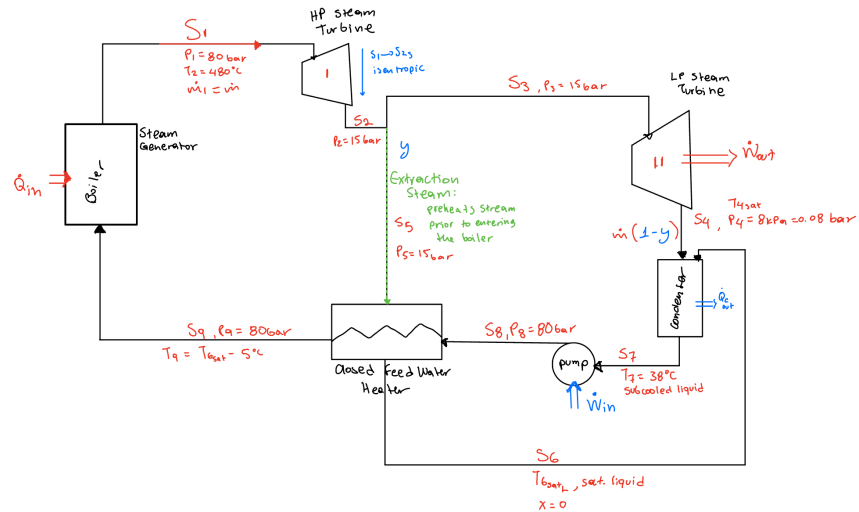


Figure 1: Schematic of regenerative Rankine cycle with closed feedwater heater.

Given Data & Constants

Table 1: Inputs and constants for the Rankine cycle analysis.

Quantity	Value
Boiler outlet pressure, P_1	80 bar
Boiler outlet temperature, T_1	480 °C
HP turbine discharge pressure	15 bar
LP turbine exhaust pressure	0.08 bar
HP turbine isentropic efficiency, $\eta_{t,HP}$	0.80
LP turbine isentropic efficiency, $\eta_{t,LP}$	0.80
Condenser outlet temperature, T_7	38 °C
Pump efficiency, η_p	0.70
Boiler efficiency, η_{boiler}	0.80
Coal lower heating value (LHV)	22 MJ/kg
CFWH shell outlet	Saturated liquid at 15 bar
Total turbine power output	1000 MW
Reference environment temperature, T_0	300 K
Heat source temperature, T_H	1300 K

3 Equipment Analysis

3.1 Pump Analysis

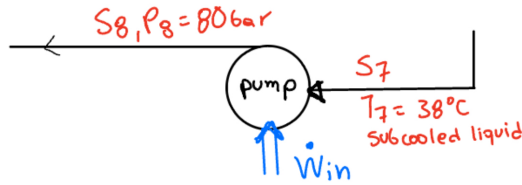


Figure 2: Schematic of the Boiler Feedwater Pump (inlet S7 and outlet S8).

ENG Calculations	Ideal Gas		
	PUMP		
Stream Properties	S7	S8rev	S8
Molar Flow, kmol/s	68.2911	68.2911	68.2911
Mass Flow, kg/s			
Volume Flow, m ³ /s			
Temperature, C	38.0000	38.2248	39.0529
Pressure, bar	0.0800	80.0000	80.0000
Vapor Fraction	0.0000	0.0000	0.0000
Molecular Weight, kg/kmol			
Molar Enthalpy, kJ/kmol	2866.084	3010.850	3072.893
Vapor Phase	0.000	0.000	0.000
Liquid Phase	2866.084	3010.850	3072.893
Molar Entropy, kJ/(kmol K)	9.824	9.824	10.023
Vapor Phase	0.000	0.000	0.000
Liquid Phase	9.824	9.824	10.023
Molar Density, kmol/m ³			
Enthalpy Flow, kJ/s	195728.18	205614.41	209851.37
Entropy Flow, kJ/(s K)	670.92	670.92	684.51
Availability, kJ/s	-5549.05	4337.18	4497.38
Duty [Power], kJ/s [kW]	-2037411.45	9886.24	14123.19
Efficiency			0.70

Figure 3: Pump Data

Energy balance:

$$-\Delta(\dot{m}h) + \dot{Q} + \dot{W}_s = 0 \quad (1)$$

Since the heat transfer is assumed negligible compared to the enthalpy of the steam, we set $\dot{Q} = 0$. Therefore, the pump is treated as adiabatic and the balance reduces to:

$$-\Delta(\dot{m}h) + \dot{W}_s = 0 \quad (2)$$

Entropy balance:

$$-\Delta(\dot{m}s) + \sum_i \frac{\dot{Q}_i}{T_i} + \dot{S}_{gen} = 0 \quad (3)$$

For an adiabatic process, $\dot{Q}_i = 0$, so this becomes:

$$-\Delta(\dot{m}s) + \dot{S}_{gen} = 0 \quad (4)$$

If the process is considered reversible, then $\dot{S}_{gen} = 0$,

$$s_7 = s_8^{rev} \quad (5)$$

For the reversible case, the pump work is:

$$\dot{W}_s^{rev} = \dot{m}(h_8^{rev} - h_7) \quad (6)$$

The lost work for the pump is expressed as:

$$LW = -\Delta\dot{m}(H - T_0S) = \dot{m}(B_7 - B_8) + \dot{W}_s,$$

The pump efficiency is defined as:

$$\eta = \frac{\dot{W}_s^{rev}}{\dot{W}_s}, \quad \eta = 0.7.$$

3.2 Closed Feed Water Heater Analysis

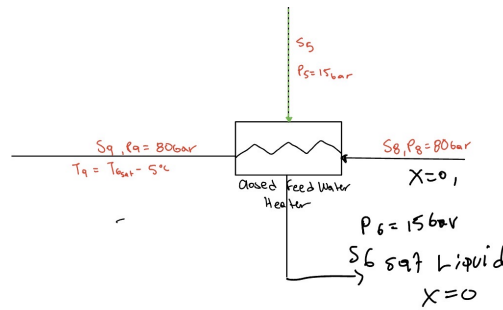


Figure 4: Schematic of the Closed Feed Water Heater (inlets S5, S8; outlets S6, S9).

ENG Calculations	Ideal Gas					
Stream Properties	S5	S6	S7	S8rev	S8	S9
Molar Flow, kmol/s	22.9830	22.9830	68.2911	68.2911	68.2911	68.2911
Mass Flow, kg/s						
Volume Flow, m ³ /s						
Temperature, °C	277.8231	198.2891	38.0000	38.2248	39.0529	193.2891
Pressure, bar	15.0000	15.0000	0.0800	80.0000	80.0000	80.0000
Vapor Fraction	1.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Molecular Weight, kg/kmol						
Molar Enthalpy, kJ/kmol	50259.952	15216.608	2866.084	3010.850	3072.893	14866.524
Vapor Phase	50259.952	0.000	0.000	0.000	0.000	0.000
Liquid Phase	0.000	15216.608	2866.084	3010.850	3072.893	14866.524
Molar Entropy, kJ/(kmol K)	123.068	41.695	9.824	9.824	10.023	40.662
Vapor Phase	123.068	0.000	0.000	0.000	0.000	0.000
Liquid Phase	0.000	41.695	9.824	9.824	10.023	40.662
Molar Density, kmol/m ³						
Enthalpy Flow, kJ/s	1155123.68	349723.06	195728.18	205614.41	209851.37	1015251.99
Entropy Flow, kJ/(s K)	2828.48	958.28	670.92	670.92	684.51	2776.88
Availability, kJ/s	306580.04	62238.54	-5549.05	4337.18	4497.38	182187.97
Duty [Power], kJ/s [kW]			-2037411.45	9886.24	14123.19	3105627.60
Efficiency					0.70	0.80

Figure 5: Closed Feed Water Heater Data

The Closed Feedwater Heater acts as a heat exchanger, with inlets S5 and S8, and outlets S6 and S9. Stream S5 is assumed to be superheated steam with vapor fraction = 1, same as S2. Stream S6 is taken as a saturated liquid at 15 bar, with vapor fraction = 0.

The energy balance for the heat exchanger is the following:

$$-\Delta\dot{m}H + \dot{Q} + \dot{W}_s = 0 \quad (7)$$

Since there is no external heat or shaft work:

$$m_5h_5 + m_8h_8 - m_6h_6 - m_9h_9 = 0 \quad (8)$$

Mass balances for hot and cold streams give:

$$m_5 = m_6, \quad m_8 = m_9$$

To determine the mass split fraction y , we entered the CFWH mass balance in Excel and used **Goal Seek** to drive the balance equation to zero by changing y . This procedure gave a split fraction of

$$y = 0.3365.$$

3.3 Turbine Analysis

3.3.1 High Pressure Turbine

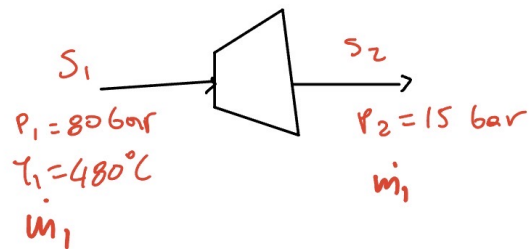


Figure 6: Schematic of the High-Pressure Turbine (HP Turbine).

ENG Calculations			
		HP TURBINE	
Stream Properties	S1	S2rev	S2
Molar Flow, kmol/s	68.2911	68.2911	68.2911
Mass Flow, kg/s			
Volume Flow, m ³ /s			
Temperature, C	480.0000	239.7419	277.8231
Pressure, bar	80.0000	15.0000	15.0000
Vapor Fraction	1.0000	1.0000	1.0000
Molecular Weight, kg/kmol			
Molar Enthalpy, kJ/kmol	60342.808	52217.517	53842.574
Vapor Phase	60342.808	52217.517	53842.574
Liquid Phase	0.000	0.000	22077.042
Molar Entropy, kJ/(kmol K)	120.011	120.011	123.068
Vapor Phase	120.011	120.011	123.068
Liquid Phase	0.000	0.000	0.000
Molar Density, kmol/m ³			
Enthalpy Flow, kJ/s	4120879.59	3565994.12	3676971.18
Entropy Flow, kJ/(s K)	8195.67	8195.67	8404.48
Availability, kJ/s	1662177.60	1107292.13	1155626.77
Duty [Power], kJ/s [kW]	3105627.60	-554885.47	-443908.41
Efficiency			0.80

Figure 7: High Pressure Turbine Data

Energy balance:

$$-\Delta(\dot{m}H) + \dot{Q} + \dot{W}_s = 0 \quad (9)$$

With $\dot{Q} = 0$:

$$-\Delta(\dot{m}H) + \dot{W}_s = 0 \quad (10)$$

Entropy balance:

$$-\Delta(\dot{m}S) + \sum_i \frac{\dot{Q}_i}{T_i} + \dot{S}_{gen} = 0 \quad (11)$$

For adiabatic turbine:

$$-\Delta(\dot{m}S) + \dot{S}_{gen} = 0 \quad (12)$$

If reversible:

$$s_1 = s_2^{rev}$$

Efficiency:

$$\eta_t = \frac{\dot{W}_s}{\dot{W}_s^{rev}}$$

Isentropic outlet ($T_{2,s}$). The reversible outlet state is obtained from the entropy balance

$$S_1 = S_{2,s}.$$

In Excel, we guess $T_{2,s}^{(0)}$ at P_2 , compute $S_{2,s}^{(0)} = \text{SSSteam}(T_{2,s}^{(0)}, P_2)$, and apply **Goal Seek** to drive $S_{2,s} - S_1 = 0$ by changing $T_{2,s}$. The corresponding enthalpy then gives

$$\dot{W}_s^{rev} = \dot{m} (H_{2,s} - H_1).$$

Actual outlet (T_2). For the irreversible state, we enforce the turbine efficiency condition

$$\frac{\dot{W}_s}{\dot{W}_s^{rev}} - 0.8 = 0.$$

We guess $T_2^{(0)}$ at P_2 , compute H_2 , and use **Goal Seek** to adjust T_2 until the above equation is satisfied, yielding (T_2, H_2, S_2) and $\dot{W}_s = \dot{m} (H_2 - H_1)$.

Lost work. Finally, the irreversibility of the HP turbine:

$$LW = \dot{m} (B_1 - B_2) + \dot{W}_s, \quad B = H - T_0 S.$$

3.3.2 Low Pressure Turbine

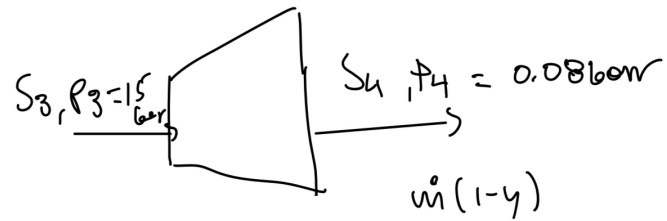


Figure 8: Schematic of the Low-Pressure Turbine (LP Turbine).

ENG Calculations		LP TURBINE	
Stream Properties	S3	S4rev	S4
Molar Flow, kmol/s	45.3082	45.3082	45.3082
Mass Flow, kg/s			
Volume Flow, m ³ /s			
Temperature, C	277.8231	41.5343	41.5343
Pressure, bar	15.0000	0.0800	0.0800
Vapor Fraction	1.0000	0.8169	0.8878
Molecular Weight, kg/kmol			
Molar Enthalpy, kJ/kmol	53842.574	38500.648	41569.034
Vapor Phase	53842.574	46426.672	46426.672
Liquid Phase	0.000	3132.177	3132.177
Molar Entropy, kJ/(kmol K)	123.068	123.068	132.819
Vapor Phase	123.068	148.256	148.256
Liquid Phase	0.000	10.675	10.675
Molar Density, kmol/m ³			
Enthalpy Flow, kJ/s	2439508.15	1744393.66	1883416.56
Entropy Flow, kJ/(s K)	5576.00	5576.00	6017.79
Availability, kJ/s	766707.37	71592.89	78080.05
Duty [Power], kJ/s [kW]		-695114.49	-556091.59
Efficiency			0.80

Figure 9: Low Pressure Turbine Data

Energy balance:

$$-\Delta(\dot{m}H) + \dot{Q} + \dot{W}_s = 0 \quad (13)$$

With $\dot{Q} = 0$:

$$-\Delta(\dot{m}H) + \dot{W}_s = 0 \quad (14)$$

Entropy balance:

$$-\Delta(\dot{m}S) + \sum_i \frac{\dot{Q}_i}{T_i} + \dot{S}_{gen} = 0 \quad (15)$$

For reversible case:

$$s_3 = s_4^{rev}$$

Efficiency:

$$\eta_t = \frac{\dot{W}_s}{\dot{W}_s^{rev}}$$

Methodology for Determining Quality at S_4^{rev} and S_4

To determine the reversible and actual outlet states of stream S_4 , we proceed as follows:

Reversible state (S_4^{rev}). For the reversible process, we use the entropy balance condition

$$S_3 = S_4^{rev}.$$

1. Guess a trial quality $x^{(0)}$ for S_4^{rev} .
2. Compute the enthalpy:

$$H_4^{rev} = x^{(0)} H_4^V + (1 - x^{(0)}) H_4^L.$$

3. Compute the entropy:

$$S_4^{rev} = x^{(0)} S_4^V + (1 - x^{(0)}) S_4^L.$$

4. Apply **Goal Seek** in Excel to adjust x until $S_4^{rev} - S_3 = 0$.
5. The resulting enthalpy gives the reversible turbine work:

$$\dot{W}_s^{rev} = \dot{m} (H_4^{rev} - H_3).$$

Actual state (S_4). For the irreversible process, we enforce the turbine efficiency relation:

$$\frac{\dot{W}_s}{\dot{W}_s^{rev}} - \eta_t = 0, \quad \eta_t = 0.8.$$

1. Guess a trial quality $x^{(0)}$ for S_4 .

2. Compute the enthalpy:

$$H_4 = x^{(0)} H_4^V + (1 - x^{(0)}) H_4^L.$$

3. Compute the turbine power:

$$\dot{W}_s = \dot{m} (H_4 - H_3).$$

4. Apply **Goal Seek** to adjust x until the efficiency condition

$$\dot{W}_s - \eta_t \dot{W}_s^{rev} = 0$$

is satisfied.

5. The final values yield the actual state (x_4, H_4, S_4) and turbine work.

3.3.3 Steam Flow Rate Calculation

The steam flow rate at S_1 is determined by enforcing that the total turbine work equals -1000 MW. In Excel, this condition was implemented using `GoalSeek`, where the molar flow rate $\dot{m}_{S1}$ was varied until the work balance was satisfied.

The resulting solution gave:

$$\dot{m}_{S1} = 68.29 \text{ kmol/s.}$$

Finally, the CHE calculator function `KMOLperSECToMMSCFD` was applied to convert the molar flow rate to gas flow units:

$$\text{Flow (MMSCFD)} = \text{KMOLperSECToMMSCFD} \times \dot{m}_{S1}.$$

Thus, the steam flow rate is:

$$\dot{m}_{S1} = 4936.27 \text{ MMSCFD.}$$

Lost work:

$$LW = \dot{m} (B_3 - B_4) + \dot{W}_s, \quad B = H - T_0 S.$$

3.4 Boiler Analysis

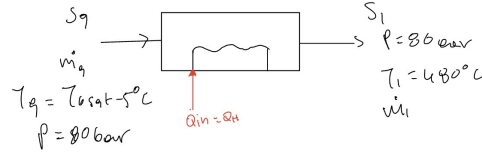


Figure 10: Schematic of the Boiler (inlet S_9 , outlet S_1).

Stream Properties	S1	S2rev	S2	S3	S4rev	S4	S5	S6	S7	S8rev	S8	S9
Molar Flow, kmol/s	68.2911	68.2911	68.2911	45.3082	45.3082	45.3082	22.9830	22.9830	68.2911	68.2911	68.2911	68.2911
Mass Flow, kg/s												
Volume Flow, m ³ /s												
Temperature, C	480.0000	238.7419	277.8231	277.8231	41.5343	41.5343	277.8231	198.2891	38.0000	38.2248	39.0529	193.2891
Pressure, bar	80.0000	15.0000	15.0000	15.0000	0.0800	0.0800	15.0000	15.0000	0.0800	80.0000	80.0000	80.0000
Vapor Fraction	1.0000	1.0000	1.0000	1.0000	0.8189	0.8878	1.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Molecular Weight, kg/kmol												
Molar Enthalpy, kJ/kmol	60342.808	52217.517	53842.574	53842.574	38500.648	41569.034	50259.952	15216.608	2866.084	3010.850	3072.893	14866.524
Vapor Phase	60342.808	52217.517	53842.574	53842.574	46426.672	46426.672	50259.952	0.000	0.000	0.000	0.000	0.000
Liquid Phase	0.000	0.000	22077.042	0.000	3132.177	3132.177	0.000	15216.608	2866.084	3010.850	3072.893	14866.524
Molar Entropy, kJ/(kmol K)	120.011	120.011	123.068	123.068	123.068	132.819	123.068	41.695	9.824	9.824	10.023	40.662
Vapor Phase	120.011	120.011	123.068	123.068	148.256	148.256	123.068	0.000	0.000	0.000	0.000	0.000
Liquid Phase	0.000	0.000	0.000	0.000	10.675	10.675	0.000	41.695	9.824	9.824	10.023	40.662
Molar Density, kmol/m ³												
Enthalpy Flow, kJ/s	4120879.59	3585994.12	3676971.18	2439508.15	1744393.66	1883416.56	1155123.68	349723.08	195728.18	205614.41	209851.37	1015251.99
Entropy Flow, kJ/(s K)	8195.67	8195.67	8404.48	5576.00	5576.00	6017.79	2828.48	958.28	670.92	670.92	694.51	2776.88
Availability, kJ/s	1662177.60	1107292.13	1155626.77	766707.37	71592.89	78080.05	306580.04	62238.54	-5549.05	4337.18	4497.38	182187.97
Duty [Power], kJ/s [kW]	3105627.60	-554885.47	-443008.41		-695114.49	-556091.59			-2037411.45	9886.24	14123.19	3105627.60
Efficiency			0.80			0.80					0.70	0.80

Figure 11: Boiler Data S9 to S1

Mass balance:

$$-\Delta \dot{m} = 0 \Rightarrow \dot{m}_g = \dot{m}_1$$

Energy balance.

$$-\Delta \dot{m} H + \dot{Q}_{in} = 0$$

$$\dot{Q}_{in} = \dot{m} \Delta H$$

$$\dot{Q}_{in} = \dot{m} (H_1 - H_9)$$

Lost work:

$$LW = -\Delta \dot{m} (H - T_0 S) + \dot{Q}_H \left(1 - \frac{T_0}{T_H} \right),$$

where the ambient temperature is $T_0 = 300$ K and the boiler source temperature is $T_H = 1300$ K.

3.5 Condenser Analysis

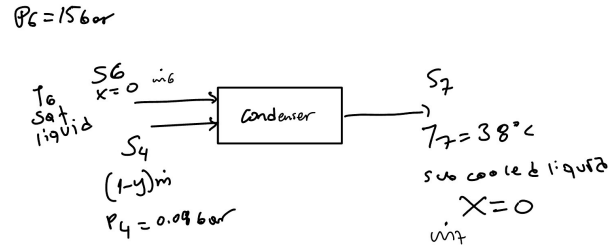


Figure 12: Schematic of the Condenser (inlets S_4, S_6 ; outlet S_7).

Stream Properties	S1	S2rev	S2	S3	S4rev	S4	S5	S6	S7	S8rev	S8	S9
Molar Flow, kmol/s	68.2911	68.2911	68.2911	45.3062	45.3062	45.3062	22.9630	22.9630	68.2911	68.2911	68.2911	68.2911
Mass Flow, kg/s												
Volume Flow, m ³ /s												
Temperature, C	480.0000	239.7419	277.8231	277.8231	41.5343	41.5343	277.8231	198.2891	38.0000	38.2248	39.0529	193.2891
Pressure, bar	80.0000	15.0000	15.0000	15.0000	0.0800	0.0800	15.0000	15.0000	0.0800	80.0000	80.0000	80.0000
Vapor Fraction	1.0000	1.0000	1.0000	1.0000	0.8169	0.8878	1.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Molecular Weight, kg/kmol												
Molar Enthalpy, kJ/kmol	60342.808	52217.517	53842.574	53842.574	38500.848	41589.034	50259.952	15216.608	2896.084	3010.850	3072.893	14866.524
Vapor Phase	60342.808	52217.517	53842.574	53842.574	46426.672	46426.672	50259.952	0.000	0.000	0.000	0.000	0.000
Liquid Phase	0.000	0.000	22077.042	0.000	3132.177	3132.177	0.000	15216.608	2896.084	3010.850	3072.893	14866.524
Molar Entropy, kJ/(kmol K)	120.011	120.011	123.068	123.068	123.068	132.819	123.068	41.695	9.624	9.624	10.023	40.662
Vapor Phase	120.011	120.011	123.068	123.068	148.256	148.256	123.068	0.000	0.000	0.000	0.000	0.000
Liquid Phase	0.000	0.000	0.000	0.000	10.675	10.675	0.000	41.695	9.624	9.624	10.023	40.662
Molar Density, kmol/m ³												
Enthalpy Flow, kJ/s	4120879.59	3565994.12	3676971.18	2439508.15	1744393.66	1883416.56	1155123.68	349723.06	195728.18	205614.41	209851.37	1015251.99
Entropy Flow, kJ/(s K)	8195.67	8195.87	8404.48	5576.00	5576.00	6017.79	2628.48	958.28	670.92	670.92	694.51	2776.88
Availability, kJ/s	1662177.60	1107292.13	1155626.77	796707.37	71592.89	78080.05	306580.04	62236.54	-5549.05	4337.19	4497.38	182187.97
Duty [Power], kJ/s [kW]	3105627.60	-554885.47	-443008.41	-695114.49	-566091.59	-566091.59	-2037411.48	9886.24	14123.19	14123.19	3105627.60	
Efficiency			0.80			0.80					0.70	0.80

Figure 13: Condenser Data S6 S4 to S7

Mass balance.

$$-\Delta \dot{m} = 0 \Rightarrow \dot{m}_6 + \dot{m}_4 = \dot{m}_7$$

Energy balance.

$$-\Delta \dot{m} H + \dot{Q}_c = 0$$

$$-\dot{m} \Delta H + \dot{Q}_c = 0$$

$$\dot{Q}_c = \dot{m}_7 H_7 - (\dot{m}_4 H_4 + \dot{m}_6 H_6)$$

Lost work:

$$LW = -\Delta \dot{m} (H - T_0 S) + \dot{Q}_c \left(1 - \frac{T_0}{T_c} \right) = \dot{m} [(B_4 + B_6) - B_7] + \dot{Q}_c \left(1 - \frac{T_0}{T_c} \right).$$

where the ambient temperature is $T_0 = 300 \text{ K}$ and the condenser temperature is $T_c = 311 \text{ K}$ (38°C).

3.6 User Calculations

ENG Calculations	
User Data and Calculations	
Entropy Balance Turbine HP:	0.0000
Efficiency Balance Turbine HP:	0.0000
Steam Flow Rate, MMSCFD	4936.2719
Lost Work HP Turbine, MW	62.6424
Entropy Balance LP Turbine	0.0000
Lost Work LP Turbine	132.5357
Efficiency Balance LP Turbine	0.0000
Power Equation MW	0.0000
Split fraction of mass flowrate y	0.3365
Energy Balance for CFWH	0.0000
Lost Work CFWH, MW	66.6509
Entropy Balance Pump	0.0000
Efficiency Balance Pump	0.0000
Lost Work Pump, MW	4.0768
Lost Work Boiler, MW	908.9547
T1(Thot), K	1300.0000
T7,(Tcold), K	311.1500
Lost Work Condenser, KW	72.8574

Figure 14: Numerical Data

3.7 Overall Cycle Equations

Total mass and energy balance.

$$\dot{Q} + \dot{W}_s = 0 \quad \Rightarrow \quad \dot{Q}_c + \dot{Q}_H + \dot{W}_{s,\text{turbine}} + \dot{W}_{s,\text{pump}} = 0$$

Thermal efficiency

$$\eta_{\text{thermal}} = \frac{\dot{W}_s}{\dot{Q}_H} = \frac{\dot{W}_{s,\text{turbine}} + \dot{W}_{s,\text{pump}}}{\dot{Q}_H}$$

Maximum thermal efficiency (Carnot)

$$\eta_{\text{max, thermal}} = 1 - \frac{T_c}{T_H}$$

Total lost work

$$LW = LW_{\text{turbine}} + LW_{\text{condenser}} + LW_{\text{pump}} + LW_{\text{boiler}} + LW_{\text{CFWH}}$$

Second-law (thermodynamic) efficiency.

$$\eta_{\text{thermodynamic}} = \frac{-\text{Goal}}{-\text{Goal} - LW}$$

Performance	
Energy Balance Check	
Thermal Efficiency	32%
Total Lost Work, MW	1247.7179
Thermodynamic Efficiency	44%
Maximun Theoretical Efficiency	76.1%
Lost in Boiler, %	72.8%
Lost in HP Turbine, %	5.0%
Lost in LP Turbine, %	10.6%
Lost in Pump, %	0.33%
Lost in Condenser, %	5.8%
Lost in CFWH, %	5.3%

Figure 15: Perfomance Calculations

4 CO₂ Emissions Estimate

The mass flow rate of coal required for the boiler is obtained from the energy input and the heating value of the fuel:

$$\dot{m}_{\text{coal}} = \frac{Q_{in}}{HV}$$

$$\dot{m}_{\text{coal}} = \frac{3105.63 \text{ MJ/s}}{22 \text{ MJ/kg}} = 141.16 \text{ kg/s coal.}$$

Assuming the coal contains 50% carbon by weight, the carbon flow rate is

$$\dot{m}_C = \dot{m}_{\text{coal}} \times 0.5 = 70.58 \text{ kg/s C.}$$

For complete combustion, all carbon is oxidized to CO₂. The mass flow rate of CO₂ is therefore found from the molar mass ratio:

$$\dot{m}_{\text{CO}_2} = \dot{m}_C \times \frac{M_{\text{CO}_2}}{M_C} = 70.58 \times \frac{44.01}{12.01} = 258.5 \text{ kg/s CO}_2.$$

To express this on an annual basis in gigatonnes per year:

$$\dot{m}_{CO_2, yr} = \dot{m}_{CO_2} \cdot \left(\frac{1 \times 10^{-12} \text{ Gt}}{\text{kg}} \right) \cdot (3.154 \times 10^7 \text{ s/yr}),$$
$$\dot{m}_{CO_2, yr} = 0.00814 \text{ Gt/yr}.$$

CO₂ Emissions Comparison: From the cycle analysis, the calculated annual emissions of CO₂ are:

$$\dot{m}_{CO_2, yr} = 0.00814 \text{ Gt/yr}.$$

For context, in 2024, energy-related CO₂ emissions worldwide reached

$$37.8 \text{ Gt/yr},$$

the highest ever recorded, according to the International Energy Agency (IEA) [1].

5 Conclusion

This study analyzed a regenerative Rankine cycle with a closed feedwater heater. Energy and entropy balances were applied to all major components (pump, turbines, boiler, condenser, and CFWH) to derive expressions for work, heat transfer, and lost work. Cycle performance was then evaluated using both First Law (thermal efficiency) and Second Law (thermodynamic efficiency) analyses.

The results show that regeneration reduces boiler heat input by preheating feedwater, improving efficiency and lowering fuel use.

Overall, the study demonstrates how regenerative cycles enhance thermal efficiency while reducing environmental impact, and emphasizes the value of combining First and Second Law methods for a complete assessment of power plant performance.

References

- [1] International Energy Agency (2025). *Global Energy Review 2025: CO₂ Emissions*. Available at: <https://www.iea.org/reports/global-energy-review-2025/co2-emissions>

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